AI Code Security Analyzer

Project Proposal

Team Name: CSE4104-7C-T04

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1. Project Title

AI Code Security Analyzer

2. Problem Statement

Modern software systems are becoming increasingly complex, yet maintaining secure coding practices remains a major challenge for developers and organizations. Software developers, students, startups, and even professional security researchers often lack access to fast, intelligent, and affordable security auditing tools capable of identifying vulnerabilities before deployment. Manual code review requires extensive cybersecurity expertise, consumes significant time, and frequently produces inconsistent results depending on the reviewer's experience and skill level.

This problem is highly important because insecure software remains one of the leading causes of cyberattacks, data breaches, ransomware incidents, and unauthorized system access worldwide. Vulnerabilities such as SQL Injection, Cross-Site Scripting (XSS), Command Injection, authentication flaws, insecure cryptographic implementations, and exposed secrets continue to compromise millions of users and cost organizations billions of dollars every year. With the rapid adoption of agile development and continuous deployment practices, security reviews are often skipped or delayed, creating large attack surfaces that attackers actively exploit.

Existing security analysis tools such as Snyk and SonarQube rely heavily on predefined rules and pattern matching techniques. Although these tools are effective at detecting known vulnerabilities, they struggle to understand application logic, contextual code flow, and complex relationships between software components. As a result, they generate excessive false positives, require complicated configuration, and fail to identify advanced logic-level vulnerabilities.

Artificial Intelligence, particularly Large Language Models (LLMs), provides a powerful solution to these limitations. Unlike traditional scanners, AI models understand source code semantically rather than purely syntactically. This allows them to detect context-dependent vulnerabilities, insecure workflows, improper authorization mechanisms, and unsafe coding practices that conventional rule-based systems frequently miss. AI can also generate human-readable explanations, map findings to industry standards such as CWE and CVSS, and provide actionable remediation guidance in real time.

To address these challenges, the proposed system aims to develop an AI-powered code security analysis platform capable of scanning source code across multiple programming languages using advanced AI providers including Claude, GPT-4o, Gemini, and DeepSeek. The platform will provide fast, intelligent, and accessible vulnerability analysis through an interactive web application designed for developers, students, and cybersecurity professionals.

3. Objectives

- Develop an intelligent AI-powered platform capable of analyzing source code for security vulnerabilities across multiple programming languages.
- Automate the process of vulnerability detection to reduce dependency on time-consuming manual code reviews.
- Integrate advanced AI providers such as Claude, GPT-4o, Gemini, and DeepSeek to improve detection accuracy, flexibility, and reliability.
- Detect and classify common and advanced security vulnerabilities including SQL Injection, Cross-Site Scripting (XSS), Command Injection, Authentication Flaws, Hardcoded Secrets, and insecure cryptographic implementations.
- Generate structured security reports containing CWE references, CVSS severity scores, vulnerability descriptions, affected code snippets, and remediation recommendations.
- Provide an interactive and user-friendly dashboard that allows users to filter and analyze security findings efficiently.
- Support direct source code input, file upload, and GitHub repository scanning for practical development workflows.
- Enable exportable reports in JSON, Markdown, and PDF formats for documentation and sharing purposes.
- Make professional-grade security analysis accessible to students, developers, startups, and small organizations without requiring advanced cybersecurity expertise.
- Improve secure software development practices by integrating intelligent AI-based security auditing into the software development lifecycle.

4. Proposed Features

- Multi-Provider AI Scanning using Claude, GPT-4o, Gemini, and DeepSeek APIs.
- Direct source code paste and secure file upload system for instant analysis.
- Public GitHub repository scanning through repository or file URL input.
- Detection of more than 50 common and advanced software vulnerabilities.
- Severity-based vulnerability filtering (Critical, High, Medium, Low).
- Interactive vulnerability reporting interface with expandable findings.

- CWE and CVSS integration for industry-standard vulnerability classification.
- AI-generated remediation suggestions and secure coding recommendations.
- Risk score dashboard with graphical visualization and severity distribution.
- Real-time scan logs displaying live analysis progress.
- Multi-language support including Python, JavaScript, TypeScript, Java, Go, PHP, Ruby, C/C++, Rust, and Kotlin.
- Downloadable reports in JSON, Markdown, and PDF formats.
- Responsive modern web interface optimized for usability and accessibility.

5. Technology Stack

Frontend

- React.js
- Vite

Backend

- FastAPI (Python)

AI Integration

- Claude API (Anthropic)
- GPT-4o API (OpenAI)
- Gemini API (Google)
- DeepSeek API

Database

- PostgreSQL
- MongoDB
- phpMyAdmin

Report Generation

- JSON
- Markdown
- PDF

Version Control

- Git
- GitHub

6. Expected Outcome

The proposed system is expected to provide developers, students, and cybersecurity professionals with a fast, intelligent, and accessible method for identifying vulnerabilities in source code without requiring advanced security expertise. Users will be able to upload or paste code and receive detailed security analysis reports within seconds, significantly reducing the time and complexity involved in manual code auditing.

The system will improve existing development workflows by simplifying security analysis and integrating it directly into the software development lifecycle. Instead of relying solely on traditional rule-based scanners or expensive enterprise security tools, users will gain access to an AI-driven platform capable of understanding code context and detecting complex vulnerabilities more effectively.

The integration of advanced AI models will enable the system to provide semantic understanding of source code, detect logic-level vulnerabilities, analyze insecure data flows, and generate human-readable explanations with remediation guidance.

Overall, the project aims to contribute toward safer software development practices by making intelligent security analysis faster, easier, and more accessible for individuals and organizations of all sizes.